

Project Workflow: Week 6

Phase: Core Logic, Arithmetic & State

Objective

Implement the core arithmetic operations of the Scientific Calculator, handle user input and operators, display calculation results, and manage the calculator state.

Workflow

1. Handle numeric input
2. Handle arithmetic operators
3. Implement arithmetic calculation logic
4. Handle the equals (=) operation
5. Manage calculator state and display results

1. Handle User Input

- Accept numeric input from the calculator buttons.
- Build the current number from the entered digits.
- Support decimal input using the decimal point button.
- Update the display whenever the input changes.

2. Handle Arithmetic Operators

- Store the first operand entered by the user.
- Store the selected arithmetic operator.
- Prepare the calculator to receive the second operand.
- Support $+$, $-$, $\times$, and $\div$ operations.

3. Implement Core Calculation Logic

- Create a calculation method for basic arithmetic operations.
- Perform addition, subtraction, multiplication, and division.
- Calculate the pending operation when required before accepting another operator.
- Keep the calculation flow connected with the display.

4. Handle Results

- Calculate the final result when the equals (=) button is pressed.
- Show the calculation history and result on the display.

- Store the latest result for later use by the calculator.
- Reset the appropriate state after displaying a result.

5. Manage Calculator State

- Maintain the current input value.
- Track the first operand and selected operator.
- Use state flags to identify new input and completed results.
- Keep the calculator ready for the next calculation.

Week 6 Deliverables

- Arithmetic operations implemented
- User input and operators handled
- Calculation logic implemented
- Results displayed correctly
- Calculator state managed
- Core arithmetic functionality completed